GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

Real-World Scale Drawings

Question Bank - Additional 5.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, use the scale given.

1. $\frac{1}{2}$ in : 1 ft . A wall is 7 in on the plan. Real length?

14 ft

1 in : 2 ft.

2. 1 in : 8 ft . A room is 1.5 in on the plan. Real length?

12 ft

3. 1 in : 8 ft . A real wall is 36 ft. Length on the plan?

4.5 in

36 ÷ 8.

4. $\frac{1}{4}$ in : 1 ft . A room is 2.5 in by 3.5 in on the plan. Real dimensions and area?

10 ft by 14 ft; 140 ft^2

1 in : 4 ft; 10 × 14.

For questions 5 through 7, answer in the unit asked.

5. A model plane is built at 1 : 60 and is 30 cm long. The real plane, in meters?

18 m

30 × 60 = 1,800 cm.

6. On a drawing a wall is 4 in long; the real wall is 22 ft. Another wall is 7 in long on the drawing. How long is it really?

38.5 ft

5.5 ft per inch; 7 × 5.5.

7. A 36-ft bus is modeled at 1 : 36 and at 1 : 54 . Which model is longer, and by how much?

The 1 : 36 model, by 4 in (12 in against 8 in)

36 ft = 432 in; 432 ÷ 36 = 12; 432 ÷ 54 = 8.

Choose one.

8. A plan is drawn at $1\text{ in} : 5\text{ ft}$. A room is 2 in by 3 in on it. What is the room's real area?

- ○ A. $25\,\text{ft}^2$
- ○ B. $30\,\text{ft}^2$
- ● **C.** $150\,\text{ft}^2$
- ○ D. $50\,\text{ft}^2$

C

10 ft by 15 ft; 10 × 15.